Paramdeep Singh Gill

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Education

Thapar Insitute of Engineering and Technology (TIET) — Patiala, Punjab
Bachelor of Engineering in Computer Engineering, CGPA: 8.09 — *Sep 2021 – Jun 2025*

Apeejay School (C.B.S.E. affliated) — Jalandhar, Punjab
10 + 2 (Non Medical), Percentage: 95.60% — *2021*

Apeejay School (C.B.S.E. affliated) — Jalandhar, Punjab
10, Percentage: 90.33% — *2019*

Experience

IIT Ropar — Rupnagar, Punjab
Research intern — *January 2025 – June 2025*

- Exploring Knowledge Distillation for Model Compression in Medical Domain. Under the guidance of Dr. Deepti Reddy Bathula (Indian Institute of Technology (IIT) Ropar, India)

AgNext — Sahibzada Ajit Singh (S.A.S.) nagar, Punjab
Android developer intern — *May 2023 – August 2023*

- Flutter Built android apps that connect to BLE devices.

Awards and Recognition

Runner-Ups - HackForSustainability — 2023
Organised by Saturnalia Fest (largest in North India) for Project Saarthi — *Patiala, Punjab*

Winner - Smart India Hackathon (SIH) (Senior Software Edition) [August 24 – 25] — Aug 2022
Team Leader and Front-End Developer — *Karnataka*

- Problem Statement under the Department of School Education and Literacy (DoSEL), Ministry of Education

Awarded Merit Scholarship – II (Tuition fee waive off) — 2022
From Thapar Insitute of Engineering and Technology — *Patiala, Punjab*

Dr Stya Paul Award for Human Values – 2020 — 2020
In Senior Secondary Category and scholarship at Apeejay School, Jalandhar — *Jalandhar, Punjab*

ATL Student Innovator Award — 2019
Certificate of Merit by Atal Innovation Mission, Niti Aayog — *Jalandhar, Punjab*

Qualified for National Championship of World Robot Olympiad (WRO) — 2016 and 2017
For the year 2016 & 2017. Secured third position in Regional Championship of WRO 2017.

On-going Projects

XAI-FL — 2026
Research project — *Remote*

- Trying to integrate XAI (Explainable Artificial Intelligence) on models trained using FL (Federated Learning). With a focus on using privacy preserving techniques like Differential Privacy.
- Guide: Dr. Parteek Kumar Bhatia (Washington State University (WSU), USA) and Dr. Sachin Kansal (TIET, India)
- Technology used: PyTorch, InterpretML, NVFlare.

BioFL — 2026
Research project — *Remote*

- Working on applying Federated Learning (FL) on unexplored biomedical datasets.
- Guide: Dr. Parteek Kumar Bhatia (WSU, USA) and Dr. Kashish Goyal (TIET, India)
- Technology used: PyTorch, InterpretML.

Deepfake Detection 2026

Group Research project *Remote*

- Working on building a model to detect deepfake in dataset having audio video modality.
- Guide: Dr. Shefali Arora Chouhan (Dr B R Ambedkar National Institute of Technology (NIT) Jalandhar, India)
- Technology used: PyTorch

Projects

ThaparSat research project 2024

Solo project *Patiala, Punjab*

- Systematic Approach to Enhance Signal Quality and Noise Reduction in Remote Sensing and Astrophysical Applications
- Capstone project (Bachelor's main project) under the guidance of Dr. Seema Bawa and Dr. Mamta Gulati (School of Mathematics)

Vriddhi App 2022

Group Project *Patiala, Punjab*

- An app to guide parents to facilitate and ensure holistic growth in cognitive, psychomotor and affective domains. Contributed in the front-end of the app (using Flutter-Dart).
- Technology used: Flutter-Dart, Firebase, GetX, React Js, Chart Js, Swiper Js, Figma.

Technical & Research Skills

Machine Learning & AI: PyTorch, InterpretML, NVFlare, OpenCV, NumPy, Matplotlib, Knowledge Distillation, Federated Learning
Programming Languages: C, C++, Python, Dart, HTML, CSS, SQL
Research & Academic Tools: LaTeX, AsciiDoc, Zotero, Git, GitHub, VS Code, JupyterLab
Development Frameworks: Django, Flutter, Firebase

Extra-Curricular and Volunteer

International Conference on Statistical Learning (ICSL) 2025

Organised by IIT Ropar *Rupnagar, Punjab*

- Presented "Systematic Approach to Enhance Signal Quality and Noise Reduction in Remote Sensing and Astrophysical Applications" i.e. the "ThaparSat research project" mentioned in the "Projects" title

Center for Training and Development (CTD) 2021 – 2022

Senior Coordinator *Patiala, Punjab*

- Coordinated the Placement Symposium, an annual event for UG final-year students to prepare them for various employability-related challenges.

Open Web Application Security Project (OWASP) 2022 - 2023

Team Member *Patiala, Punjab*

- Delivered a talk on "Blockchain Technology" during the Orientation Session.

International Award for Young People (IAYP) 2019

Successfully completed Bronze Category Course *Jalandhar, Punjab*

Prefectorial Board of School 2019 – 2021

Captain and Vice-Captain of Cleanliness *Jalandhar Punjab*

References

Dr. Deepti Reddy Bathula +91 95011 96606, bathula@iitrpr.ac.in

Associate Professor, IIT Ropar *Rupnagar, Punjab*

Dr. Seema Bawa +91 98156 04297, seema@thapar.edu

Professor, TIET *Patiala, Punjab*

Dr. Shefali Arora Chouhan +91 98888 13400, chouhansa@nitj.ac.in

Assistant Professor (Grade-II), NIT Jalandhar *Jalandhar, Punjab*